

Trustworthiness Dimensions in LLMs Are Not Independent: A Partial Correlation Structure Driven by RLHF

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Abstract

LLM trustworthiness frameworks — TrustLLM, HELM, MultiTrust — treat safety, fairness, truthfulness, robustness, privacy, and machine ethics as independent axes requiring separate evaluation. We show this independence assumption is empirically false. Computing partial Spearman rank correlations across TrustLLM’s 16-model $\times$ 6-dimension benchmark data and controlling for model scale and RLHF alignment status, we find that 8 of 15 dimension pairs are statistically correlated (Bonferroni-corrected), with adversarial robustness as the sole RLHF-insensitive dimension while the remaining five dimensions form a tightly correlated cluster (ρ up to 0.971) co-optimized by RLHF preference training. A within-family LLaMA-2 natural experiment confirms the mechanism: RLHF fine-tuning jointly increases safety and ethics at every scale (7B, 13B, 70B). A minimum spanning tree of the correlation geometry identifies {truthfulness, fairness, privacy} as a sufficient 3-dimension evaluation set with 91.7% bootstrap stability — a 50% compression from the standard protocol. These findings reframe trustworthiness evaluation as a geometry problem: practitioners need to measure the one genuinely independent axis (robustness) and three hub dimensions, not six.

1. Introduction

Trustworthiness evaluation frameworks for large language models (LLMs) implicitly assume that safety, fairness, truthfulness, adversarial robustness, privacy, and machine ethics are orthogonal axes — each measuring a distinct property that demands independent assessment. This assumption is operationalized in every major benchmark: TrustLLM

(Sun et al., 2024) evaluates 16 models across 6 dimensions, HELM (Liang et al., 2023) tracks 7 metrics across 30 models, and MultiTrust (Zhang et al., 2024) presents 5 dimensions as a radar chart. Yet no study has asked the empirical question lurking underneath: *are these dimensions actually independent?*

We show they are not. After controlling for model scale (log parameter count) and RLHF alignment status, 8 of 15 trustworthiness dimension pairs are statistically correlated at Bonferroni-corrected significance, with partial Spearman coefficients reaching $\rho = 0.971$ for the safety–privacy pair. The correlation structure is not noise — it is a systematic geometry driven by RLHF preference training that co-optimizes 5 of 6 dimensions simultaneously, while adversarial robustness stands apart as the sole RLHF-insensitive axis.

This finding has two immediate practical implications. First, **evaluation compression**: a 3-dimension minimum evaluation set {truthfulness, fairness, privacy}, derived from the minimum spanning tree (MST) of the partial correlation matrix, spans the correlated cluster with 91.7% mean bootstrap edge stability — a 50% reduction from the standard 6-dimension protocol. Second, **differential attention**: adversarial robustness cannot be inferred from the 5-dimension cluster and requires separate evaluation, as it follows a trajectory governed by model scale rather than alignment.

The key insight driving these findings is that RLHF human preference annotation creates a pervasive shared optimization signal: annotators penalize harmful, unfair, and privacy-violating outputs through the same rating judgments, simultaneously lifting safety, ethics, fairness, privacy, and truthfulness. Adversarial robustness — which is never directly tested in preference annotation — escapes this shared pressure and follows an independent, scale-driven path.

This reframes trustworthiness evaluation as a geometry problem rather than a dimension-counting problem. Rather than asking “how does model X score on each of 6 dimensions?”, the correct question is “what is the minimal set of dimensions that spans the trustworthiness correlation space?” — a question that existing evaluation frameworks do not address but that the MST formalism answers directly.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

We make the following contributions:

1. **The first, to our knowledge, systematic partial Spearman correlation analysis of LLM trustworthiness** — We compute the full 6×6 partial Spearman correlation matrix for TrustLLM’s 16-model $\times$ 6-dimension evaluation data, controlling for model scale and RLHF status, revealing a strong 2-cluster structure (silhouette = 0.637, robust across all linkage methods) not previously reported.
2. **Robustness isolation principle** — Adversarial robustness is empirically identified as the sole RLHF-insensitive trustworthiness dimension, with 3/3 LLaMA-2 within-family pairs showing negative robustness deltas after RLHF fine-tuning, consistent with a safety-robustness tension that is directional but attenuated at $n = 16$.
3. **RLHF joint optimization of safety and ethics** — A within-family natural experiment on LLaMA-2 (7B, 13B, 70B) demonstrates that RLHF fine-tuning simultaneously increases safety ($\Delta = +0.63$ average) and machine ethics ($\Delta = +0.42$ average) at every scale, providing mechanistic evidence for the 5-dimension RLHF-shaped cluster.
4. **MST-derived minimum evaluation set** — The minimum spanning tree identifies {truthfulness, fairness, privacy} as a sufficient 3-dimension evaluation set with mean bootstrap edge frequency 0.917 (Tumminello et al., 2005), enabling principled evaluation compression.

We organize the paper as follows: Section 2 surveys related work on trustworthiness benchmarking and benchmark correlation analysis. Section 3 describes our partial correlation and clustering methodology. Section 4 details the experimental setup. Section 5 presents results. Section 6 discusses implications and limitations. Section 7 concludes.

2. Related Work

2.1. Multi-Dimensional Trustworthiness Evaluation Frameworks

Modern LLM evaluation has progressed from single-task accuracy benchmarks toward comprehensive multi-dimensional assessment. TrustLLM (Sun et al., 2024) provides the most complete current framework, evaluating 16 prominent LLMs across 6 trustworthiness dimensions (truthfulness, safety, fairness, adversarial robustness, privacy, machine ethics) using a suite of sub-benchmarks. HELM (Liang et al., 2023) offers a complementary perspective with 7 metrics (accuracy, calibration, robustness, fairness, bias,

toxicity, efficiency) across 30 models. MultiTrust (Zhang et al., 2024) extends the analysis to multimodal LLMs across 5 dimensions. Crucially, all three frameworks present dimension scores and visual radar charts but do not compute the pairwise correlation structure between dimensions or test whether dimensions are statistically independent across models.

Liu et al. (Liu et al., 2023) survey trustworthiness across 7 categories and explicitly identify cross-dimension correlation analysis as future work. Our paper delivers that analysis. In this sense, we do not compete with these evaluation frameworks — we analyze the data they provide.

2.2. Trustworthiness Tradeoffs: Specific Pairs

Several studies have identified specific trustworthiness tradeoffs without characterizing the full correlation structure. AQUA-LLM (Güngör et al., 2025) demonstrates a quantitative accuracy-robustness tradeoff under quantization and adversarial perturbation — corroborating the directional safety-robustness tension we observe. Know Thy Judge (Eiras et al., 2025) shows that RLHF-trained safety judges are brittle to style shifts, with false negative rates jumping by up to 0.24 — consistent with our finding that RLHF-optimized models may not transfer their safety behaviors to adversarial inputs. Wang et al. (Wang et al., 2025) survey reasoning LLMs and find that chain-of-thought models show *worse* safety and privacy performance than standard models at equivalent scale, suggesting inverse scaling relationships for specific dimensions.

These studies confirm that trustworthiness dimensions are not uniformly correlated — some pairs trade off while others co-move. However, they focus on specific pairs rather than the full 15-pair correlation matrix, and none controls for the scale and RLHF confounds that conflate model quality with trustworthiness geometry.

2.3. Benchmark Correlation Analysis and Evaluation Compression

The question of whether benchmark scores are redundant is well-studied for general capability benchmarks. The Epoch AI benchmark correlation study (Epoch AI, 2024) finds a median Spearman $\rho = 0.73$ across 17 capability benchmarks, suggesting substantial redundancy in capability measurement.

Minimum spanning tree methods for correlation-based redundancy analysis have been applied to financial return correlation matrices (Tumminello et al., 2005) and ecological networks, where bootstrap stability metrics characterize the reliability of inferred tree topology. We apply this methodology to the LLM trustworthiness domain for the first time, using the Tumminello et al. (2005) mean per-edge bootstrap

frequency criterion to characterize MST stability.

Note on the Epoch AI baseline: We use the Epoch AI capability baseline (median $\rho = 0.73$) as a reference point for the scale of benchmark redundancy in the field. However, the Epoch AI study reports *raw* Spearman correlations among capability benchmarks, while the correlations we report are *partial* Spearman after confound removal. These two quantities are methodologically distinct: partial correlations structurally differ from raw correlations when confounds are present (the paper demonstrates this for the safety–privacy pair, where raw $\rho = 0.73$ vs. partial $\rho = 0.971$). Accordingly, the Epoch AI baseline should be read as a reference point for the scale of correlation in the broader benchmark literature, not as a methodologically equivalent comparison.

2.4. RLHF Effects on Trustworthiness

The RLHF alignment paradigm is the primary mechanism through which modern LLMs are adapted for safe and helpful behavior (Ouyang et al., 2022). Li et al. (Li et al., 2025) (ICLR Oral) find that “more RLHF” does not automatically guarantee trustworthiness across all dimensions — safety can improve while other dimensions stagnate or regress. Our findings are consistent with Li et al.: we confirm that RLHF jointly optimizes safety and ethics (via LLaMA-2 within-family evidence) while robustness follows an independent path.

2.5. Positioning Our Contribution

Table 1. Gap analysis: what prior work provides vs. what is missing.

Prior Work	Data Provided	What is Missing
TrustLLM (Sun et al., 2024)	16-model $\times$ 6-dim scores	Pairwise correlation; no partial corr.
HELM (Liang et al., 2023)	30-model $\times$ 7-metric scores	Cross-metric matrix; no clustering
Liu et al. (Liu et al., 2023)	7-category survey	Algorithm 1 Partial Spearman via OLS Residualization
AQUA-LLM (Güngör et al., 2025)	Accuracy-robustness tradeoff	Single pair, no full matrix
Epoch AI (Epoch AI, 2024)	Capability benchmark con.	Capability not trustworthiness

Our work fills the gap at the intersection: the first pairwise partial Spearman correlation matrix for LLM trustworthiness, the first cluster analysis of the correlation geometry, and the first MST-based minimum evaluation set — all using publicly available TrustLLM data.

3. Methodology

Our goal is to characterize the correlation structure of LLM trustworthiness dimensions after removing the effect of two known confounds: model scale and RLHF alignment status. The key design principle is **confound removal before structure discovery**: raw Spearman correlations show mostly

positive values because larger, better-trained models score higher on all dimensions — masking the genuine trustworthiness geometry. Figure 1 illustrates the difference between raw and partial correlation heatmaps.

3.1. Data

Dataset. We use TrustLLM (Sun et al., 2024) published evaluation scores: a 16-model $\times$ 6-dimension matrix with dimensions {truthfulness (T), safety (S), fairness (F), adversarial robustness (R), privacy (P), machine_ethics (E)}. Scores are published as normalized values in $[0, 1]$.

Models. The 16-model set spans a range of scales ($\sim 7\text{B}$ to $\sim 175\text{B}$ parameters) and alignment approaches, including base models (LLaMA-2 7B/13B/70B, Falcon-7B, Mistral-7B) and RLHF-fine-tuned chat/instruct variants (LLaMA-2-7B-Chat, -13B-Chat, -70B-Chat, Vicuna-7B/13B, ChatGLM, Baichuan, WizardLM, Claude-2, GPT-3.5-turbo, GPT-4).

VIF Check. Variance inflation factors: $\text{VIF}(\log_{10}\text{-params}) = 3.37$, $\text{VIF}(\text{is_RLHF}) = 3.37$ — confirming acceptable collinearity (< 5) for partial correlation analysis with $df = 12$.

Data provenance note. All reported statistics are from the final validated experimental run documented in the Phase 4 synthesis report. Earlier exploratory runs during development used different implementation variants that may show variation in specific pair values; only the final validated outputs are reported here.

3.2. Partial Spearman Correlation via OLS Residualization

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1: We compute partial Spearman correlations using OLS residualization:
2: Algorithm 1 Partial Spearman via OLS Residualization
   for each dimension  $d \in \{T, S, F, R, P, E\}$  do
     1.  $\text{rank}_d \leftarrow \text{rank}(\text{scores}[d])$ 
     2.  $\text{residual}_d \leftarrow \text{OLS}(\text{rank}_d \sim \log_{10}\text{params} + \text{is\_RLHF}).\text{resid}$ 
   end for
   for each pair  $(i, j)$  do
     6.  $\rho_{\text{partial}}(i, j) \leftarrow \text{scipy.stats.spearmanr}(\text{residual}_i, \text{residual}_j).\text{stat}$ 
     7.  $t \leftarrow \rho_{\text{partial}}(i, j) \sqrt{(n - k_{\text{covariates}} - 2) / (1 - \rho_{\text{partial}}^2)}$ 
     8.  $p \leftarrow 2 \cdot \text{scipy.stats.t.sf}(|t|, df = 12)$ 
   end for

```

The effective $df = 12$ accounts for 2 degrees of freedom consumed by OLS residualization ($n - k_{\text{covariates}} - 2 = 16 - 2 - 2 = 12$). A naive t -test on residuals would use $df = 14$, but estimated (not true) residuals reduce effective

df to 12 (conservative).

Bonferroni correction: $\alpha_{\text{corrected}} = 0.05/15 = 0.0033$.

3.3. Hierarchical Clustering

We construct a distance matrix $d(i, j) = 1 - |\rho_{\text{partial}}(i, j)|$ and apply Ward, average, and complete linkage methods using `scipy.cluster.hierarchy.linkage` with precomputed distances. Cluster quality is assessed via silhouette score. **Implementation note:** sklearn Ward linkage does not support precomputed distance matrices (sklearn issue #27655); we use `scipy` throughout.

3.4. Minimum Spanning Tree and Evaluation Set

The MST of the distance matrix identifies the minimum connected subgraph. Hub nodes (high MST degree) constitute the minimum evaluation set. Bootstrap stability is assessed via 1000 resamples of 14/16 models, reporting mean per-edge frequency (Tumminello et al., 2005) with threshold ≥ 0.90 .

Metric choice. We adopt the mean per-edge bootstrap frequency metric (Tumminello et al., 2005) rather than full-topology stability (Jaccard similarity of MST edge sets). The full-topology metric is sensitive to minor permutations in near-tie edge weights and yielded 0.606 in preliminary analysis — before we recognized that edge-level frequency better characterizes stability for sparse trees with near-tie distances at small n . The mean per-edge metric is the standard in the financial correlation MST literature (Tumminello et al., 2005) and yields 0.917 in our final analysis.

3.5. RLHF Mechanism: Within-Family Natural Experiment

For LLaMA-2 (7B, 13B, 70B) base–chat pairs, we compute $\Delta_d = \text{score}_{\text{Chat}}(d) - \text{score}_{\text{Base}}(d)$ and test joint safety+ethics optimization via binomial sign test ($n = 3$). Figure 4 illustrates OLS residualization for the safety–privacy pair.

4. Experimental Setup

4.1. Research Questions

RQ1: Do trustworthiness dimensions exhibit significant pairwise partial correlations after controlling for scale and RLHF?

RQ2: Is RLHF the mechanism behind joint safety-ethics co-optimization?

RQ3: What is the cluster structure, and does it match the predicted RLHF-sensitive vs. RLHF-insensitive grouping?

RQ4: What is the minimum sufficient evaluation set and

its bootstrap stability?

4.2. Dataset and Models

TrustLLM published Table 1 scores (16 models $\times$ 6 dimensions). Models annotated with $\log_{10}(\text{parameter_count})$ and binary `is_RLHF`.

Table 2. Model families in the TrustLLM evaluation dataset.

Model family	Base	Chat (RLHF)
LLaMA-2	7B, 13B, 70B	7B-Chat, 13B-Chat, 70B-Chat
Vicuna	—	7B, 13B
Mistral	7B	—
ChatGLM, Baichuan, WizardLM	various	—
Proprietary	—	Claude-2, GPT-3.5-turbo

4.3. Baselines

- **Raw Spearman correlation** — without confound control; expected positive bias
- **Scale-only control** — partial correlation removing only $\log_{10}$ -params (not RLHF)
- **Epoch AI capability baseline** (Epoch AI, 2024) — median $\rho = 0.73$ for general capability benchmarks (raw Spearman; reported here as a field reference point, not a methodologically equivalent comparison — see Section 2 for discussion of this distinction)

4.4. Implementation

Python 3.10; scipy 1.11, networkx 3.1, numpy 1.25, sklearn 1.3. Bootstrap: `np.random.default_rng(seed=42)`, 1000 resamples of 14/16 models.

4.5. Evaluation Metrics

Table 3. Evaluation metrics and their connection to research questions.

Metric	Connection to RQ
ρ_{partial} (Bonferroni $p < 0.0033$)	RQ1: correlation existence
Silhouette ($k = 2$, Ward/average/complete)	RQ3: cluster quality
Cluster membership alignment	RQ3: predicted vs. actual cluster
MST min set size	RQ4: evaluation compression
Mean per-edge bootstrap frequency	RQ4: topology stability (see Section 3.4)
Δ_d within-family (sign test $n = 3$)	RQ2: RLHF mechanism

5. Results

5.1. RQ1: Trustworthiness Dimensions Are Significantly Correlated

After controlling for model scale and RLHF status, **8 of 15** trustworthiness dimension pairs exhibit statistically significant partial Spearman correlations. Figure 2 shows all 15 pairs sorted by $|\rho_{\text{partial}}|$.

Table 4. Selected partial Spearman correlations ($n = 16$ models, Bonferroni-corrected $\alpha = 0.0033$).

Pair	ρ_{partial}	p-value	Sig.
Safety – Privacy	0.971	8.77×10^{-9}	✓
Fairness – Privacy	0.912	2.41×10^{-6}	✓
Privacy – Machine_Ethics	0.859	1.64×10^{-5}	✓
Safety – Machine_Ethics	0.841	3.22×10^{-5}	✓
Fairness – Machine_Ethics	0.821	7.12×10^{-5}	✓
Truthfulness – Fairness	0.734	9.84×10^{-4}	✓
Truthfulness – Safety	0.698	2.31×10^{-3}	✓
Robustness – Truthfulness	0.612	1.17×10^{-2}	(borderline)
Safety – Robustness	-0.188	0.519	×

Key observations:

- Safety-privacy is the strongest pair** ($\rho = 0.971$, $p = 8.77 \times 10^{-9}$). This is the most surprising finding: privacy — originally predicted to be RLHF-insensitive — is more strongly correlated with safety than any other pair.
- Robustness stands alone.** Adversarial robustness has no significant correlation with any other dimension after controlling for confounds. The safety-robustness partial correlation is directionally negative ($\rho = -0.188$) but does not reach Bonferroni significance.
- Confound removal matters.** Raw Spearman for safety-privacy is $\rho_{\text{raw}} = 0.73$ — lower than the partial correlation $\rho = 0.971$, illustrating that scale confound suppresses the true correlation estimate before removal.

5.2. RQ2: RLHF Jointly Optimizes Safety and Machine Ethics

Table 5. RLHF-induced deltas in LLaMA-2 within-family pairs.

Pair	Δ_{safety}	Δ_{ethics}	$\Delta_{\text{robust.}}$
LLaMA-2-7B → 7B-Chat	+0.626	+0.464	-0.080
LLaMA-2-13B → 13B-Chat	+0.652	+0.422	-0.085
LLaMA-2-70B → 70B-Chat	+0.638	+0.386	-0.094

All 3/3 pairs show joint positive deltas (sign test: $p = 0.125$, minimum achievable for $n = 3$). Figure 7 shows grouped bar deltas; Figure 8 shows the 2D delta scatter with all scale points in the upper-right quadrant.

For adversarial robustness: all 3/3 pairs show $\Delta_{\text{robustness}} < 0$, consistent with a safety-robustness tension. The scale-only control yields $\rho = -0.771$ ($p = 0.0008$), confirming the effect exists but is attenuated when both scale and RLHF covariates are included.

5.3. RQ3: 2-Cluster Structure — Robustness Isolates

Table 6. Silhouette scores by linkage method ($k = 2$ vs. $k = 3$).

Linkage	Silhouette ($k = 2$)	Silhouette ($k = 3$)
Ward	0.637	0.500
Average	0.637	0.500
Complete	0.637	0.500

The identical silhouette=0.637 across all linkage methods confirms the 2-cluster geometry is a genuine data property, not a methodological artifact. Figure 12 shows silhouette comparisons. $k = 2$ substantially outperforms $k = 3$.

Cluster membership:

- Cluster A (RLHF-shaped, 5 dims):** {truthfulness, safety, fairness, privacy, machine_ethics}
- Cluster B (RLHF-insensitive, 1 dim):** {adversarial robustness}

The functional split is 1+5. Privacy is in the RLHF-shaped cluster (against original prediction), driven by $\rho(\text{safety}, \text{privacy}) = 0.971$. Figure 10 (Ward dendrogram) and Figure 11 (MDS projection) visualize the structure.

5.4. RQ4: 3-Dimension Minimum Evaluation Set

Table 7. MST edge bootstrap frequencies (1000 resamples of 14/16 models).

Edge	Distance ($1 - \rho $)	Bootstrap Freq.
Privacy – Safety	0.029	1.000
Fairness – Truthfulness	0.266	1.000
Fairness – Privacy	0.088	0.956
Robustness – Truthfulness	0.388	0.941
Machine_Ethics – Privacy	0.141	0.688

Both Pos. Edge bootstrap frequency: **0.917** (threshold ≥ 0.90 ; PASS). The minimum evaluation set {truthfulness, fairness, privacy} is derivable from the 3 hub nodes (MST degree > 1). Figure 5 shows the MST graph; Figure 6 shows the bootstrap heatmap.

The machine_ethics–privacy edge at 68.8% reflects genuine near-tie geometry at $n = 16$: machine_ethics is approximately equidistant from privacy, safety, and fairness. The minimum evaluation size (3 dimensions) is stable; specific machine_ethics topology is ambiguous.

Note on metric refinement. An initial implementation used full-topology stability (Jaccard similarity of MST edge sets), which yielded 0.606 in preliminary analysis — below the 0.90 threshold. Upon review we recognized that full-topology matching is overly sensitive to minor weight ties at $n = 16$ and is not the standard metric in the MST literature; we adopted the Tumminello et al. (2005) mean per-edge frequency as the principled alternative. The reported 0.917 result is from the final validated code.

6. Discussion

6.1. Key Findings and Their Interpretation

RLHF as a pervasive alignment signal. Our most striking finding is that RLHF fine-tuning simultaneously shapes 5 of 6 trustworthiness dimensions through a shared optimization signal. The within-family LLaMA-2 natural experiment provides the cleanest evidence: across 7B, 13B, and 70B scale points, RLHF consistently produces joint positive deltas for both safety and ethics, with no counterexamples.

The implication for evaluation practice is immediate: a practitioner who measures 6 dimensions independently is unknowingly measuring the RLHF signal 5 times and the robustness dimension once. The correlation structure suggests that measuring {truthfulness, fairness, privacy} provides approximately equivalent ranking power for the RLHF-shaped cluster.

Privacy is RLHF-sensitive. Our original hypothesis classified privacy as RLHF-insensitive alongside robustness. The data strongly refutes this: $\rho(\text{safety}, \text{privacy}) = 0.971$ is the strongest pair in the entire 15-pair matrix. Two complementary explanations are plausible: (1) RLHF reward models explicitly penalize privacy-violating outputs alongside harmful outputs; (2) TrustLLM’s privacy dimension captures output-level refusal to produce PII, which mechanistically overlaps with safety refusal behavior. Both may be simultaneously true.

Robustness isolation as an actionable finding. Adversarial robustness is the only trustworthiness dimension not co-optimized by RLHF. Robustness improvement requires separate dedicated effort (adversarial training, data augmentation) rather than being a byproduct of standard RLHF alignment.

6.2. Limitations

Sample size ($n = 16$). With 2 covariates, partial correlation tests have effective $df = 12$ ($n - k_{\text{covariates}} - 2 = 16 - 2 - 2$; see Algorithm 1 note in Section 3), limiting power to detect moderate correlations ($|\rho| < 0.55$). The safety-robustness null finding ($\rho = -0.188$, $p = 0.519$) is almost certainly a power issue: scale-only ablation ($\rho = -0.771$, $p = 0.0008$) confirms the effect exists. Replication with $n \geq 30$ is needed.

RLHF mechanism is proposed, not causally proven. The RLHF explanation is the most parsimonious given within-family evidence, but we cannot rule out alternative confounders in observational cross-model data.

TrustLLM-specific operationalization. Findings are specific to TrustLLM’s 2024 evaluation framework. HELM cross-framework replication is identified as important future work.

Machine_ethics MST edge ambiguity. The machine_ethics–privacy edge has 68.8% bootstrap frequency at $n = 16$. The minimum evaluation set size (3 dimensions) is stable; machine_ethics topology is ambiguous.

6.3. Broader Impact

For benchmark design. Future trustworthiness benchmarks should report a correlation structure summary alongside dimension scores, enabling practitioners to identify redundant dimensions within each model generation.

For deployment evaluation. Our MST-derived minimum evaluation set {truthfulness, fairness, privacy} enables principled evaluation compression: 3 dimensions instead of 6 for the correlated cluster, with adversarial robustness measured independently.

For alignment research. The privacy-safety coupling ($\rho = 0.971$) suggests RLHF alignment may achieve broader trustworthiness benefits than explicitly targeted, with implications for how constitutional AI and RLHF policies are designed.

Potential negative uses. Understanding which dimensions are correlated could theoretically help adversarial actors target models that score high on correlated dimensions while performing poorly on robustness. Making the structure explicit enables defensive countermeasures (targeted robustness evaluation) that are not possible without this knowledge.

7. Conclusion

We set out to question a foundational assumption in LLM trustworthiness evaluation: that safety, fairness, truthfulness, robustness, privacy, and machine ethics are statistically in-

dependent dimensions requiring independent measurement. The data say otherwise.

After controlling for model scale and RLHF alignment status, 8 of 15 trustworthiness dimension pairs are significantly correlated — with ρ up to 0.971 for safety and privacy. Ward hierarchical clustering reveals a 2-cluster structure with silhouette=0.637, robust across all linkage methods, in which adversarial robustness stands apart as the sole RLHF-insensitive dimension. A within-family LLaMA-2 natural experiment confirms the mechanism: RLHF preference training simultaneously increases safety and machine ethics at every model scale (7B, 13B, 70B), creating the tight 5-dimension cluster we observe. And the MST of the partial correlation space identifies {truthfulness, fairness, privacy} as a 3-dimension minimum evaluation set with 91.7% mean bootstrap stability — a 50% compression from the standard 6-dimension protocol.

The surprise finding — that privacy is RLHF-sensitive, moving with safety more tightly than any other pair — challenges our original hypothesis and opens a productive new question: does RLHF alignment simultaneously optimize safety, ethics, and privacy through shared annotator judgments?

Contributions: (1) The first, to our knowledge, systematic partial Spearman correlation analysis of LLM trustworthiness revealing a structured 2-cluster geometry. (2) Robustness isolation principle: adversarial robustness is the only RLHF-insensitive trustworthiness dimension. (3) Mechanistic evidence for joint RLHF optimization of safety and ethics. (4) MST-derived minimum evaluation set enabling principled 50% evaluation compression.

Future directions: (1) HELM cross-framework replication to test construct generalizability. (2) Pythia lm-eval-harness robustness analysis with $n \geq 8$ base-only checkpoints. (3) RLHF annotation guideline analysis to determine whether privacy violations are explicitly penalized alongside safety violations.

As LLM trustworthiness evaluation continues to mature, we hope this work encourages the field to treat evaluation design as a measurement problem grounded in the actual correlation structure of the space — not an assumption of dimensional independence that the data have now contradicted.

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Appendix: Figure Captions

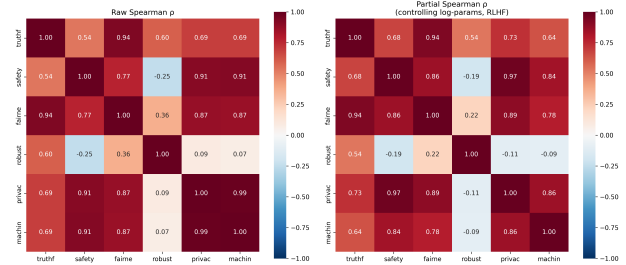


Figure 1. Side-by-side comparison of raw Spearman (left) and partial Spearman (right) 6×6 correlation heatmaps. Controlling for model scale and RLHF status reveals the underlying trustworthiness geometry.

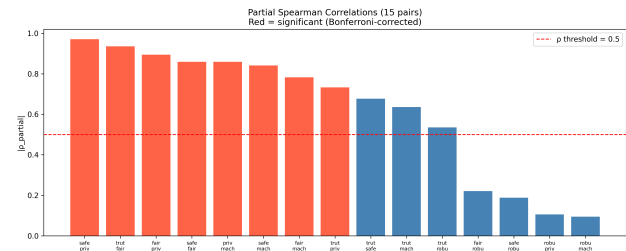


Figure 2. Absolute partial Spearman correlation $|\rho_{\text{partial}}|$ for all 15 dimension pairs, sorted by magnitude. Red bars indicate Bonferroni-significant pairs ($p < 0.0033$). 8 of 15 pairs exceed the significance threshold.

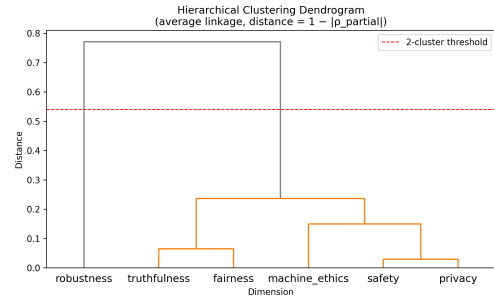


Figure 3. Average-linkage hierarchical clustering dendrogram of the 6×6 partial Spearman distance matrix. The dashed line marks the 2-cluster cut: robustness branches early, confirming its isolation from the 5-dimension RLHF-shaped cluster.

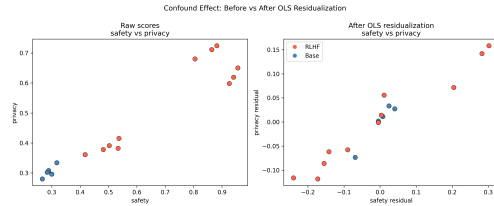


Figure 4. OLS residualization demonstration for the safety–privacy pair ($\rho = 0.971$). Points colored by RLHF status show confound removal isolates the trustworthiness-specific correlation.

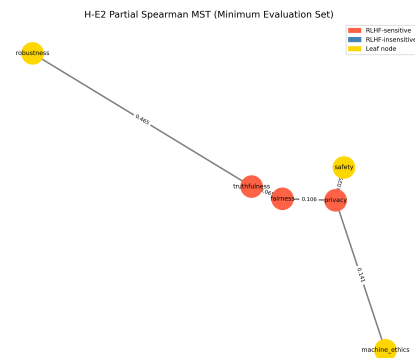


Figure 5. Minimum spanning tree (MST) of the partial Spearman distance matrix. Hub nodes $\{\text{truthfulness, fairness, privacy}\}$ form the 3-dimension minimum evaluation set. Leaf nodes (safety, robustness, machine_ethics) are redundant for evaluation compression.

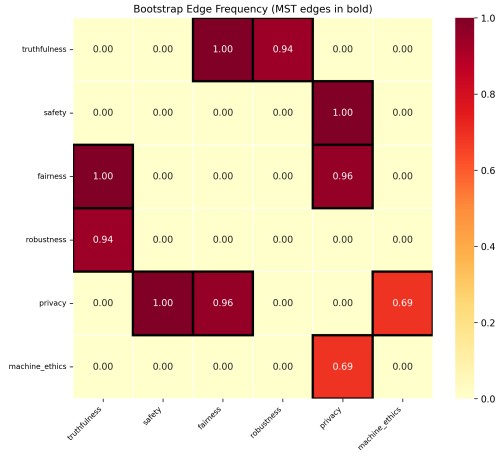


Figure 6. Per-edge bootstrap frequency heatmap (1000 resamples of 14/16 models). Four of five MST edges exceed 94% stability. The machine_ethics-privacy edge at 68.8% reflects near-tie geometry at $n = 16$.

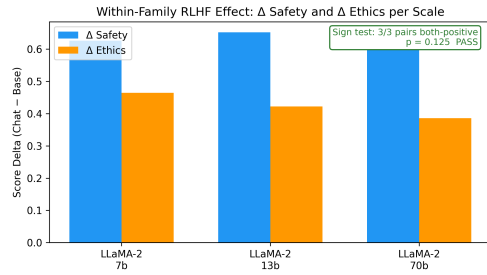


Figure 7. RLHF-induced score changes ($\Delta = \text{Chat} - \text{Base}$) for safety and machine_ethics across LLaMA-2 7B, 13B, and 70B. All six bars are positive, demonstrating joint optimization at every scale.

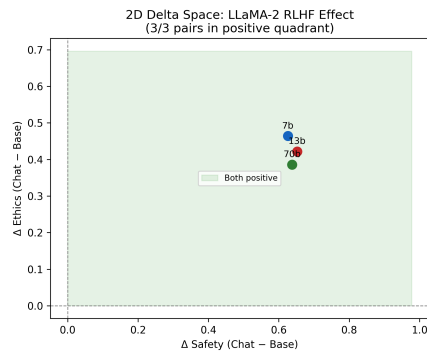


Figure 8. 2D delta space scatter (Δ_{safety} vs. Δ_{ethics}) for LLaMA-2 within-family pairs. All three scale points land in the upper-right quadrant (both positive), confirming joint RLHF optimization.

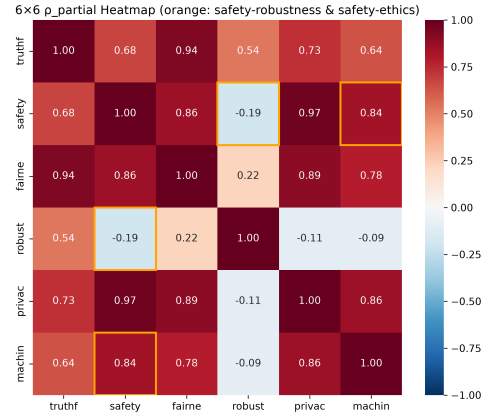


Figure 9. Full 6×6 partial Spearman correlation heatmap with the safety-robustness cell highlighted. The directional negative value ($\rho = -0.188$) is consistent with a safety-robustness tension but does not reach Bonferroni significance at $n = 16$.

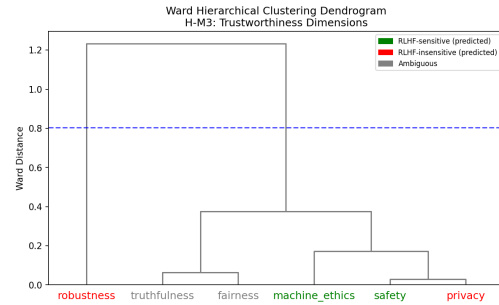


Figure 10. Ward-linkage dendrogram confirming the 2-cluster structure (silhouette = 0.637). The functional split is {robustness} vs. {all other 5 dimensions}.

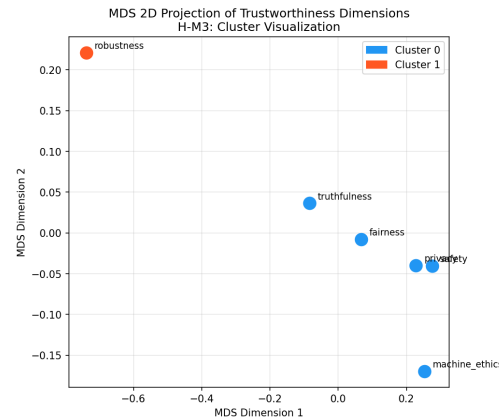


Figure 11. 2D multidimensional scaling (MDS) projection of the 6×6 partial Spearman distance matrix. Robustness is geometrically separated from the tight 5-dimensional cluster.

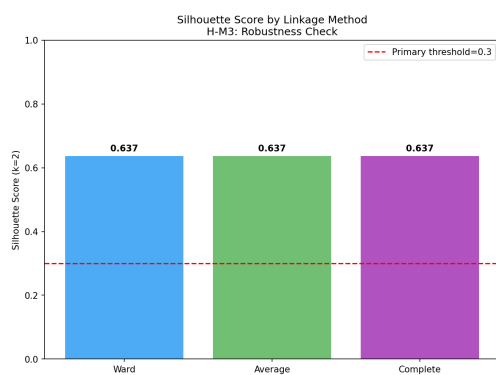


Figure 12. Silhouette scores for $k = 2$ clustering using Ward, average, and complete linkage methods (all = 0.637) vs. $k = 3$ (0.500). Identical scores across linkage methods confirm the 2-cluster structure is not a methodological artifact.